

Classifying Clothing Images with a CNN

*Final Assignment, Deep Learning and Reinforcement Learning course, IBM Machine Learning Professional Certificate
Dataset: Fashion-MNIST (public image classification benchmark)*

Main Objective

This project builds an image classifier using a convolutional neural network (CNN). The task is multi-class classification: given a small grayscale image of a clothing item, predict which of 10 categories it belongs to.

Three CNN variations are trained and compared under the same conditions, and the best one is recommended based on test accuracy and training behavior, not accuracy alone.

Dataset Description

The dataset is Fashion-MNIST, a public benchmark of 70,000 grayscale images, each 28 by 28 pixels, split into 10 clothing categories: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle boot. It was built as a slightly harder, more realistic drop-in replacement for the classic MNIST digit dataset.

54,000 images were used for training, 6,000 were held out for validation, and the standard 10,000-image test set was used only for the final evaluation, never during training or model selection. Each image is a single input made of 784 pixel values (28x28), normalized to a 0 to 1 range before training. There are no other features: the raw pixel grid is what the CNN learns from.

Data Exploration and Preparation

Fashion-MNIST ships pre-cleaned: no missing values, no corrupted images, and the classes are perfectly balanced (7,000 images per category). The only preparation needed was converting pixel values to tensors, scaling them to 0 to 1, and splitting off a validation set from the training data (90/10) so model selection would not depend on the test set.

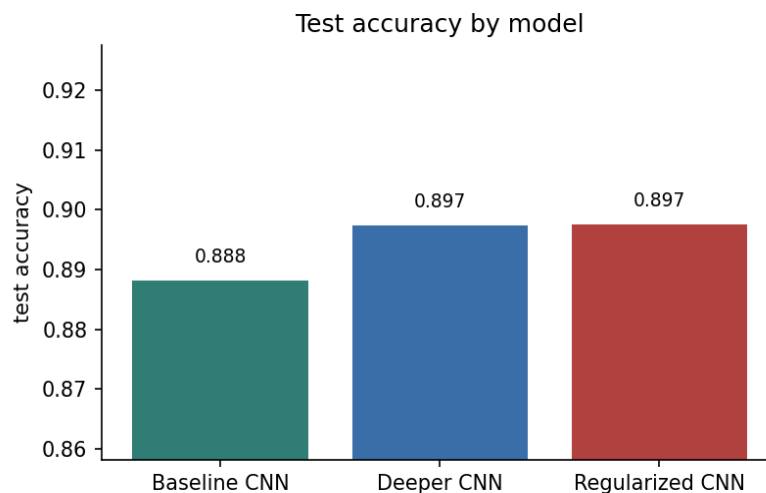
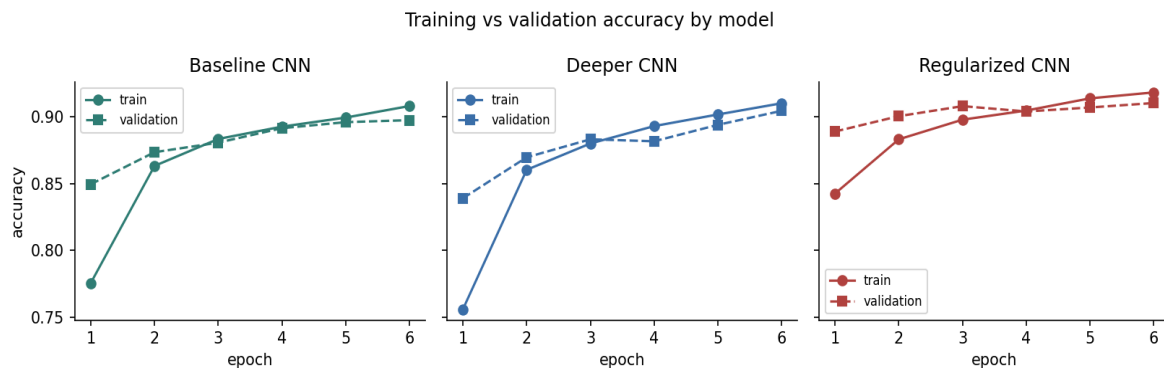
Model Variations

Three CNN architectures were trained for 6 epochs each, with the same optimizer (Adam) and batch size (128), so the comparison is fair.

- **Baseline CNN:** 2 convolutional layers (16 and 32 filters), no regularization. 206,922 parameters.
- **Deeper CNN:** 3 convolutional layers (32, 64, 64 filters), no regularization. 130,890 parameters, fewer than the baseline despite more layers, because the extra pooling step shrinks the feature map before it reaches the fully connected layers.
- **Regularized CNN:** same 2-layer structure as the baseline, plus batch normalization after each convolution and a dropout layer before the final classifier. 207,018 parameters.

Model	Test accuracy	Training time
Baseline CNN	88.82%	77s
Deeper CNN	89.74%	144s

Regularized CNN	89.75%	100s
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Final Model Recommendation

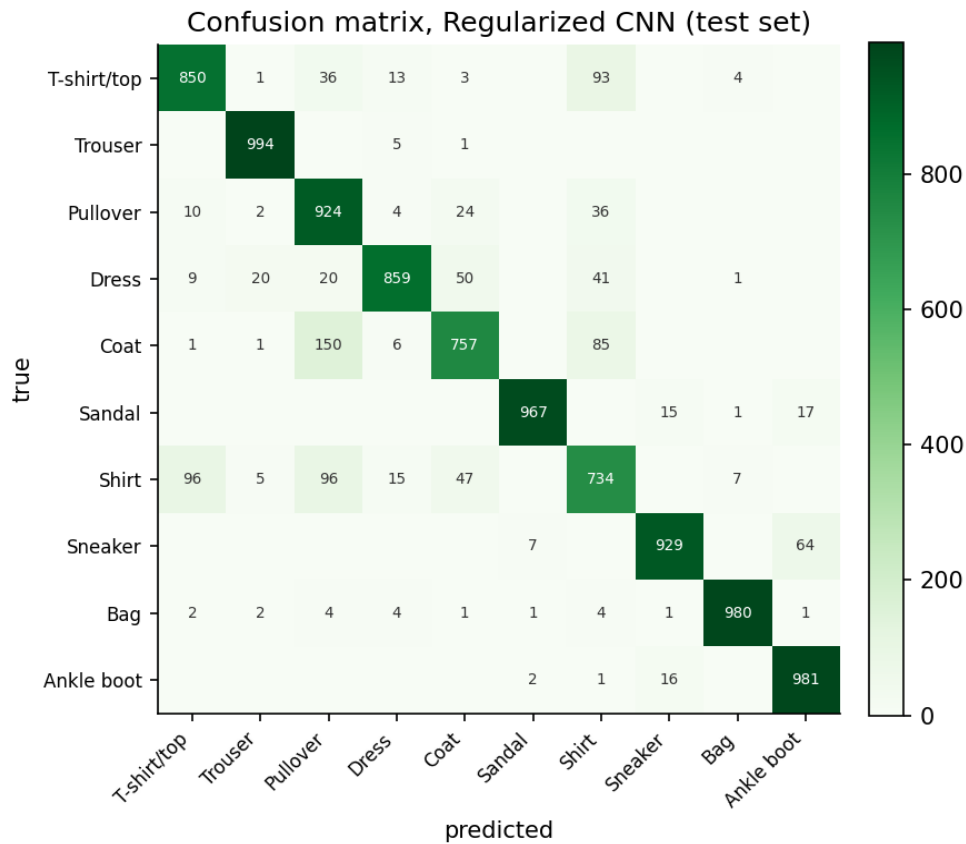
The Regularized CNN is recommended. It reaches essentially the same test accuracy as the Deeper CNN (89.75% versus 89.74%, not a meaningful difference), but trains about 30% faster and converges much sooner: it already reaches 88.9% validation accuracy after a single epoch, while the other two models need 3 to 4 epochs to get there. Batch normalization is doing real work here, not just adding parameters for nothing.

Adding depth (the Deeper CNN) gave almost the same benefit as adding regularization (the Regularized CNN) over the baseline, but cost nearly twice the training time for no real accuracy gain. On CPU, where training time actually matters, that is a worse trade.

Key Findings and Insights

- The 10 classes are not equally hard. Trouser (99.4%), Bag (98.0%), and Ankle boot (98.1%) are almost perfectly classified. Shirt (73.4%) and Coat (75.7%) are the hardest by far.

- The confusion matrix explains why. Shirt gets confused with T-shirt/top, Pullover, and Coat; Coat gets confused with Pullover and Shirt. These are all upper-body garments with a similar silhouette at 28x28 resolution, with no color or texture to tell them apart. A person looking at the same tiny grayscale thumbnails would likely make some of the same mistakes.
- In the Regularized CNN's training curve, validation accuracy sits slightly above training accuracy for most epochs. That is not a bug: dropout is only active during training, so the model scores a bit lower on the data it is training on than it does at evaluation time, when dropout is turned off.



Limitations

- Only 6 training epochs were run per model, mainly to keep training time reasonable on CPU. Accuracy was still improving slowly at epoch 6 for all three models, so none of them had fully converged.
- Fashion-MNIST images are tiny (28x28) and grayscale. Real product photos have color, texture, and background clutter that this dataset does not capture, so this accuracy will not transfer directly to a real-world clothing classifier.
- No data augmentation was used (no rotation, flipping, or cropping), which usually helps CNNs generalize better, especially on visually similar classes like Shirt and Coat.
- Only one random seed was tested per model. The 0.01 percentage point gap between the Deeper and Regularized CNN is within normal run-to-run noise and could flip with a different seed.

Next Steps and Future Work

- Train the Regularized CNN for more epochs with early stopping, to see its actual ceiling instead of stopping at a fixed 6.
- Add data augmentation, since the main errors are concentrated in a handful of visually similar classes that more varied training examples might help separate.
- Combine depth and regularization in one model, since the two seemed to help independently in this comparison.
- Test on a small set of real clothing photos, resized to grayscale 28x28, to see how much accuracy drops outside of the clean benchmark.